Data Envelopment Analysis

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Managerial motivation

Introduction: Managerial motivation

- In this chapter we compare the **efficiency** of similar units
- Problem class: **LP**, one per unit
- The dual says whom to learn from and by how much to reduce resources

Twelve branches to compare

- A bank has twelve branches: they use staff and operating costs, and produce loans and deposits
- Which ones perform well? By how much should the others reduce their resources?
- The outputs have no common price, and a large branch is not inefficient just because it spends more
- We need a comparison with the **best observed practices**

Data Envelopment Analysis

- **DEA** (*Data Envelopment Analysis*) builds an efficiency index by solving one LP per unit
- No price fixed in advance: each unit chooses the most favourable weights
- Used for hospitals, schools, public offices, plants, departments

In this chapter we will learn how to

1. define relative efficiency as a weighted ratio of outputs to inputs
2. turn the fractional model into an LP and write its dual
3. read in the dual the reference units and the improvement targets
4. separate technical inefficiency from scale inefficiency

Summary: Managerial motivation

- Similar units with several inputs and several outputs
- No common price for the outputs
- One LP per unit: efficiency, peers, targets

Efficiency and the CCR model

Introduction: Efficiency and the CCR model

- We define the units, the data and relative efficiency
- We write the fractional model of Charnes, Cooper and Rhodes (CCR)
- We turn it into an LP: the multiplier form

Data (input of the model)

The units are called DMUs (*Decision Making Units*)

Symbol	Type	Meaning
n	$\in \mathbb{Z}_{\geq 1}$	number of DMUs, indexed by $j \in \{1, 2, \ldots, n\}$
m	$\in \mathbb{Z}_{\geq 1}$	number of inputs, indexed by $i \in \{1, 2, \ldots, m\}$
p	$\in \mathbb{Z}_{\geq 1}$	number of outputs, indexed by $r \in \{1, 2, \ldots, p\}$
x_{ij}	$\in \mathbb{Q}_{>0}$	amount of input i used by DMU j
y_{rj}	$\in \mathbb{Q}_{>0}$	amount of output r produced by DMU j
o	$\in \{1, 2, \ldots, n\}$	the DMU being evaluated

Relative efficiency

Definition 1: Relative efficiency

Given weights $u_r \geq 0$ *and* $v_i \geq 0$, *not all zero, the efficiency ratio of DMU* j *is*

$$E_j(u, v) = \frac{\sum_{r=1}^{p} u_r \, y_{rj}}{\sum_{i=1}^{m} v_i \, x_{ij}}.$$

The relative efficiency of DMU o *is the maximum of* $E_o(u, v)$ *over the weights for which no DMU has a ratio greater than* 1. *DMU* o *is efficient if it equals* 1.

The problem in words

- *We decide* the weights of the inputs and of the outputs
- *The objective* is to maximise the efficiency ratio of DMU o
- *The constraints*: with the same weights no DMU has a ratio greater than 1
- We introduce the following $p + m$ non-negative variables:

$$\begin{cases} u_r = \text{weight of output } r, & \forall r \in \{1, 2, \ldots, p\}, \\ v_i = \text{weight of input } i, & \forall i \in \{1, 2, \ldots, m\}. \end{cases}$$

The fractional model

Fractional CCR model for DMU o

$$\max \quad \frac{\sum_{r=1}^{p} u_r\, y_{ro}}{\sum_{i=1}^{m} v_i\, x_{io}} \tag{12.1a}$$

$$\text{subject to} \quad \frac{\sum_{r=1}^{p} u_r\, y_{rj}}{\sum_{i=1}^{m} v_i\, x_{ij}} \leq 1, \quad \forall j \in \{1, 2, \ldots, n\}, \tag{12.1b}$$

$$u_r \geq 0, \quad \forall r \in \{1, 2, \ldots, p\}, \tag{12.1c}$$

$$v_i \geq 0, \quad \forall i \in \{1, 2, \ldots, m\}. \tag{12.1d}$$

- (12.1a) maximises the ratio of DMU o
- (12.1b): no DMU exceeds the ratio 1 (n fractional constraints)
- (12.1c) and (12.1d) define the variables

The Charnes–Cooper transformation

- If all the weights are multiplied by the same $t > 0$, no ratio changes
- We choose t so that the denominator of the objective equals 1:

$$\sum_{i=1}^{m} v_i \, x_{io} = 1$$

- The objective becomes the numerator alone
- Each constraint (12.1b), multiplied by its positive denominator, becomes linear

The multiplier form

CCR model, multiplier form, for DMU o

$$\max \quad \sum_{r=1}^{p} u_r \, y_{ro} \tag{12.2a}$$

$$\text{subject to} \quad \sum_{i=1}^{m} v_i \, x_{io} = 1, \tag{12.2b}$$

$$\sum_{r=1}^{p} u_r \, y_{rj} - \sum_{i=1}^{m} v_i \, x_{ij} \leq 0, \quad \forall j \in \{1, 2, \ldots, n\}, \tag{12.2c}$$

$$u_r \geq 0, \quad \forall r \in \{1, 2, \ldots, p\}, \tag{12.2d}$$

$$v_i \geq 0, \quad \forall i \in \{1, 2, \ldots, m\}. \tag{12.2e}$$

Objective function and constraints

- (12.2a) maximises the weighted output of DMU o: it is its ratio, because the weighted input equals 1
- (12.2b) normalises the weighted input of DMU o to 1 (1 linear constraint)
- (12.2c): for every DMU the weighted output does not exceed the weighted input (n linear constraints)
- (12.2d) and (12.2e) define the variables
- The optimal value $\tilde{h}_o \in (0, 1]$ is the relative efficiency; for n DMUs we solve n LPs

Summary: Efficiency and the CCR model

- Efficiency = weighted output/input ratio, with the most favourable weights
- The fractional model is invariant to scaling the weights
- Fixing the weighted input to 1 gives an LP

The envelopment form: the dual

Introduction: The envelopment form

- We write the dual of the multiplier form
- We read it as a virtual DMU built from the observed DMUs
- We obtain peers, targets and shadow prices

The dual variables

- The primal is a maximisation: one equality constraint and $n \leq$ constraints
- To follow the DEA literature the dual variables are not called π:

$$\begin{cases} \theta = \text{dual variable of constraint (12.2b)}, \\ \lambda_j = \text{dual variable of constraint (12.2c) of DMU } j. \end{cases}$$

- θ is free (equality), $\lambda_j \geq 0$ ($\leq$ constraints of a maximisation)
- Full theory: Duality theory (Linear Programming course)

The envelopment form

CCR model, envelopment form, for DMU o

$$\min \quad \theta \tag{12.3a}$$

$$\text{subject to} \quad \theta\, x_{io} - \sum_{j=1}^{n} \lambda_j\, x_{ij} \geq 0, \qquad \forall i \in \{1, 2, \ldots, m\}, \tag{12.3b}$$

$$\sum_{j=1}^{n} \lambda_j\, y_{rj} \geq y_{ro}, \quad \forall r \in \{1, 2, \ldots, p\}, \tag{12.3c}$$

$$\theta \gtreqless 0, \tag{12.3d}$$

$$\lambda_j \geq 0, \qquad \forall j \in \{1, 2, \ldots, n\}. \tag{12.3e}$$

Objective function and constraints

- (12.3a) minimises the factor θ by which all the inputs of DMU o are contracted
- (12.3b): the combination weighted by the λ_j uses at most θ times each input of o (m constraints, from the v_i)
- (12.3c): the same combination produces at least each output of o (p constraints, from the u_r)
- (12.3d) and (12.3e) define the variables

What the envelopment form says

- $\sum_{j=1}^{n} \lambda_j \left(x_{\cdot j}, y_{\cdot j}\right)$ is a **virtual** DMU
- With $\lambda_o = 1$ and the other $\lambda_j = 0$ we get $\theta = 1$: hence $\tilde{\theta} \leq 1$
- Strong duality: $\tilde{\theta} = \tilde{h}_o$
- The optimal radial contraction of the inputs **is** the relative efficiency

Three pieces of managerial information

- **Peers**: the reference set $\{j : \tilde{\lambda}_j > 0\}$, the efficient DMUs to learn from
- **Targets**: the projection of o onto the efficient frontier
- **Shadow prices** of constraints (12.3b) and (12.3c): they are the optimal weights $\tilde{v}_i$ and $\tilde{u}_r$ of the primal (one of the optimal vectors, if it is not unique)

Summary: The envelopment form

- The dual looks for the virtual DMU that does at least as much with less
- $\tilde{\theta} = \tilde{h}_o$ by strong duality
- The $\tilde{\lambda}_j > 0$ identify the peers

BCC, scale and slacks

Introduction: BCC, scale and slacks

- Variable returns to scale: the BCC model
- Scale efficiency as the CCR/BCC ratio
- Slacks, targets, and weak and strong efficiency

Returns to scale

- The CCR model assumes **constant** returns to scale: double inputs, double outputs
- For branches of very different sizes the assumption is strong
- The model of Banker, Charnes and Cooper (BCC) requires $\sum_{j=1}^{n} \lambda_j = 1$
- The virtual DMU becomes a **convex** combination, of comparable size

The BCC model

BCC model, envelopment form, for DMU o

$$\min \quad \theta \tag{12.4a}$$

$$\text{subject to} \quad \theta\, x_{io} - \sum_{j=1}^{n} \lambda_j\, x_{ij} \geq 0, \qquad \forall i \in \{1, 2, \ldots, m\}, \tag{12.4b}$$

$$\sum_{j=1}^{n} \lambda_j\, y_{rj} \geq y_{ro}, \quad \forall r \in \{1, 2, \ldots, p\}, \tag{12.4c}$$

$$\sum_{j=1}^{n} \lambda_j = 1, \tag{12.4d}$$

$$\theta \gtreqless 0, \tag{12.4e}$$

$$\lambda_j \geq 0, \qquad \forall j \in \{1, 2, \ldots, n\}. \tag{12.4f}$$

Scale efficiency

- (12.4d) shrinks the region: in a minimisation $\tilde{\theta}^{\mathrm{BCC}} \geq \tilde{\theta}^{\mathrm{CCR}}$

Definition 2: Scale efficiency

$$\mathrm{SE}_o = \frac{\tilde{\theta}_o^{\mathrm{CCR}}}{\tilde{\theta}_o^{\mathrm{BCC}}} \in (0, 1].$$

It equals 1 *at the most productive scale; it is smaller than* 1 *when part of the inefficiency depends only on size.*

- In the CCR: $\sum_{j=1}^{n} \tilde{\lambda}_j < 1$ too small, > 1 too large (if the solution is unique)

Slacks: the second phase

Second phase: maximum slacks for DMU o

$$\max \quad \sum_{i=1}^{m} s_i^- + \sum_{r=1}^{p} s_r^+ \tag{12.5a}$$

$$\text{subject to} \quad \sum_{j=1}^{n} \lambda_j x_{ij} + s_i^- = \tilde{\theta} x_{io}, \quad \forall i \in \{1, 2, \ldots, m\}, \tag{12.5b}$$

$$\sum_{j=1}^{n} \lambda_j y_{rj} - s_r^+ = y_{ro}, \quad \forall r \in \{1, 2, \ldots, p\}, \tag{12.5c}$$

$$\lambda_j \geq 0, \quad \forall j \in \{1, 2, \ldots, n\}, \tag{12.5d}$$

$$s_i^- \geq 0, \quad \forall i \in \{1, 2, \ldots, m\}, \tag{12.5e}$$

$$s_r^+ \geq 0, \quad \forall r \in \{1, 2, \ldots, p\}. \tag{12.5f}$$

Targets and weak and strong efficiency

- Targets of DMU o:

$$\hat{x}_{io} = \tilde{\theta}\, x_{io} - \tilde{s}_i^-, \qquad \hat{y}_{ro} = y_{ro} + \tilde{s}_r^+$$

Definition 3: Weak and strong efficiency

DMU o is strongly efficient if $\tilde{\theta} = 1$ and all the optimal slacks are zero; it is weakly efficient if $\tilde{\theta} = 1$ but at least one slack is positive.

Summary: BCC, scale and slacks

- BCC: $\sum_{j=1}^{n} \lambda_j = 1$, variable returns
- $\mathrm{SE} = \tilde{\theta}^{\mathrm{CCR}} / \tilde{\theta}^{\mathrm{BCC}}$
- The second phase measures the slacks and gives the targets

Worked example by hand

Introduction: Worked example by hand

- Five DMUs with one input and one output
- The solver gives the values, then we check them
- CCR, duality, BCC and weak efficiency on the same example

The instance

DMU	1	2	3	4	5
x_{1j}	2	4	6	5	2
y_{1j}	2	6	7	3	1

Multiplier form for DMU 1

$$
\begin{aligned}
\max \quad & 2\,u_1 & & & & \\
\text{subject to} \quad & & 2\,v_1 & = & 1 \\
& 2\,u_1 & - 2\,v_1 & \leq & 0 \\
& 6\,u_1 & - 4\,v_1 & \leq & 0 \\
& 7\,u_1 & - 6\,v_1 & \leq & 0 \\
& 3\,u_1 & - 5\,v_1 & \leq & 0 \\
& u_1 & - 2\,v_1 & \leq & 0 \\
& u_1, & v_1 & \geq & 0
\end{aligned}
$$

Example (1/3)

- **Step 1 — the solver values.**

$$\tilde{\theta}^{\mathrm{CCR}} = (0.6667,\ 1,\ 0.7778,\ 0.4,\ 0.3333)$$

$$\tilde{\theta}^{\mathrm{BCC}} = (1,\ 1,\ 1,\ 0.5,\ 1)$$

DMU 1: $\tilde{u}_1 = 0.3333$, $\tilde{v}_1 = 0.5$, only peer DMU 2 with $\tilde{\lambda}_2 = 0.3333$; DMU 5 (BCC): slack $\tilde{s}_1^+ = 1$

- **Step 2 — check of the CCR.** The ratio y/x divided by the best one, 1.5:

$$\tilde{h}_1 = \frac{2/2}{1.5} = 0.6667, \quad \tilde{h}_3 = \frac{7/6}{1.5} = 0.7778, \quad \tilde{h}_4 = \frac{3/5}{1.5} = 0.4$$

Example (2/3)

- **Step 3 — duality for DMU 1.** Normalisation, efficiency, active constraint of DMU 2:

$$2 \cdot 0.5 = 1, \qquad 2 \cdot 0.3333 = 0.6667, \qquad 6 \cdot 0.3333 - 4 \cdot 0.5 = 0$$

The virtual DMU $\tilde{\lambda}_2 \cdot (4, 6)$:

$$0.3333 \cdot 4 = 1.3333 = 0.6667 \cdot 2, \qquad 0.3333 \cdot 6 = 2$$

Example (3/3)

- **Step 4 — BCC for DMU 4.** $\tilde{\lambda}_1 = 0.75$, $\tilde{\lambda}_2 = 0.25$:

  $$0.75 \cdot 2 + 0.25 \cdot 4 = 2.5 = 0.5 \cdot 5, \qquad 0.75 \cdot 2 + 0.25 \cdot 6 = 3$$

  Scale efficiency: DMU 4 0.8, DMU 1 0.6667, DMU 3 0.7778

- **Step 5 — weak efficiency.** DMU 5 has $\tilde{\theta}^{\mathrm{BCC}} = 1$, but DMU 1 uses the same input and produces 1 more: with $\tilde{s}_1^+ = 1$ DMU 5 is weakly efficient

Summary: Worked example by hand

- With one input and one output the CCR efficiency is (y/x) divided by the best one
- The active constraint of the primal identifies the peer of the dual
- The BCC compares with convex combinations

Case study and results

Introduction: Case study and results

- Twelve branches, two inputs and two outputs
- CCR, BCC and scale efficiencies, peers and targets
- The frontier in a one-input, one-output view

The branch data

branch	1	2	3	4	5	6	7	8	9	10	11	12
staff (FTE)	10	25	23	18	17	27	9	24	12	10	20	30
costs (k€)	720	1420	1130	940	990	1620	640	1300	690	500	1020	1370
loans (M€)	19.6	60.7	43.5	31.7	39.6	39.7	21.9	34.5	29.2	23.7	42.8	63.5
deposits (M€)	18.5	58.7	63.0	50.5	43.2	73.1	38.0	72.7	28.7	26.3	45.7	62.9

- Realistic synthetic data generated by the script with a fixed seed (`dea_filiali.csv`)
- FTE: *Full-Time Equivalent*

The model in Gurobi

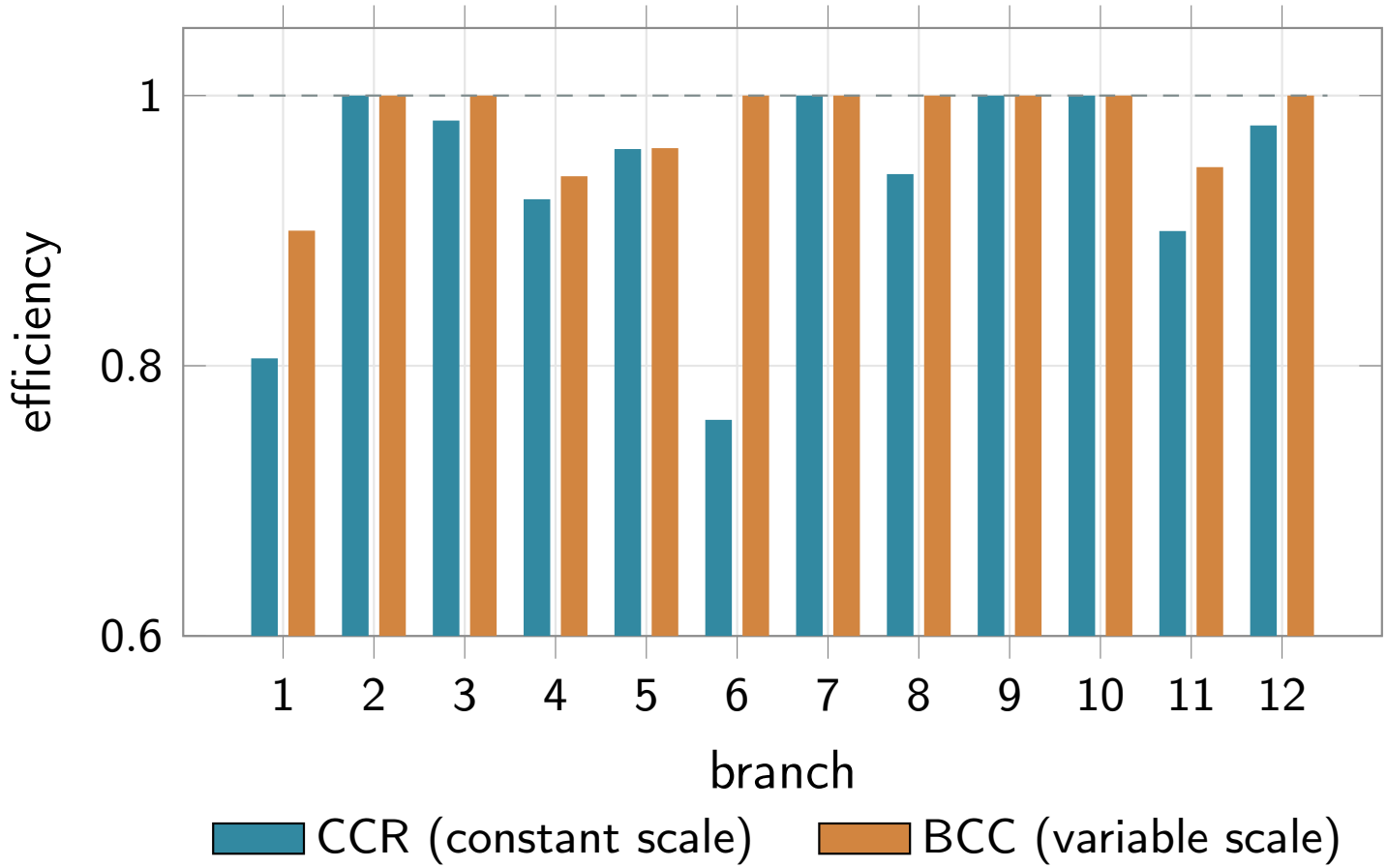
```
def inviluppo(X, Y, o, vrs=False):
    m_, n_ = X.shape           # inputs x DMUs
    p_ = Y.shape[0]            # outputs
    md = gp.Model("dea_envelopment")
    theta = md.addVar(lb=-GRB.INFINITY, name="theta")
    lam = md.addVars(n_, name="lambda")
    c_in = md.addConstrs(
        (theta * X[i, o] - gp.quicksum(lam[j] * X[i, j]
         for j in range(n_)) >= 0 for i in range(m_)), name="input")
    c_out = md.addConstrs(
        (gp.quicksum(lam[j] * Y[r, j] for j in range(n_))
         >= Y[r, o] for r in range(p_)), name="output")
    if vrs:                    # BCC: convex combination
        md.addConstr(lam.sum() == 1, name="convexity")
    md.setObjective(theta, GRB.MINIMIZE)
    md.optimize()
    return md, theta, lam, c_in, c_out
```

Results

branch	CCR	BCC	scale	sum lambda (CCR)	returns
1	0.8055	0.9000	0.8950	0.671	increasing
2	1.0000	1.0000	1.0000	1.000	constant
3	0.9815	1.0000	0.9815	1.917	decreasing
4	0.9233	0.9403	0.9819	1.423	decreasing
5	0.9604	0.9611	0.9993	1.450	decreasing
6	0.7600	1.0000	0.7600	1.924	decreasing
7	1.0000	1.0000	1.0000	1.000	constant
8	0.9419	1.0000	0.9419	1.913	decreasing
9	1.0000	1.0000	1.0000	1.000	constant
10	1.0000	1.0000	1.0000	1.000	constant
11	0.8997	0.9470	0.9501	1.641	decreasing
12	0.9779	1.0000	0.9779	2.679	decreasing

- For every branch $\tilde{\theta}$ (dual) $= \tilde{h}_o$ (primal)

CCR and BCC efficiencies



- CCR-efficient: 2, 7, 9, 10; BCC-efficient also 3, 6, 8, 12

Peers and input targets

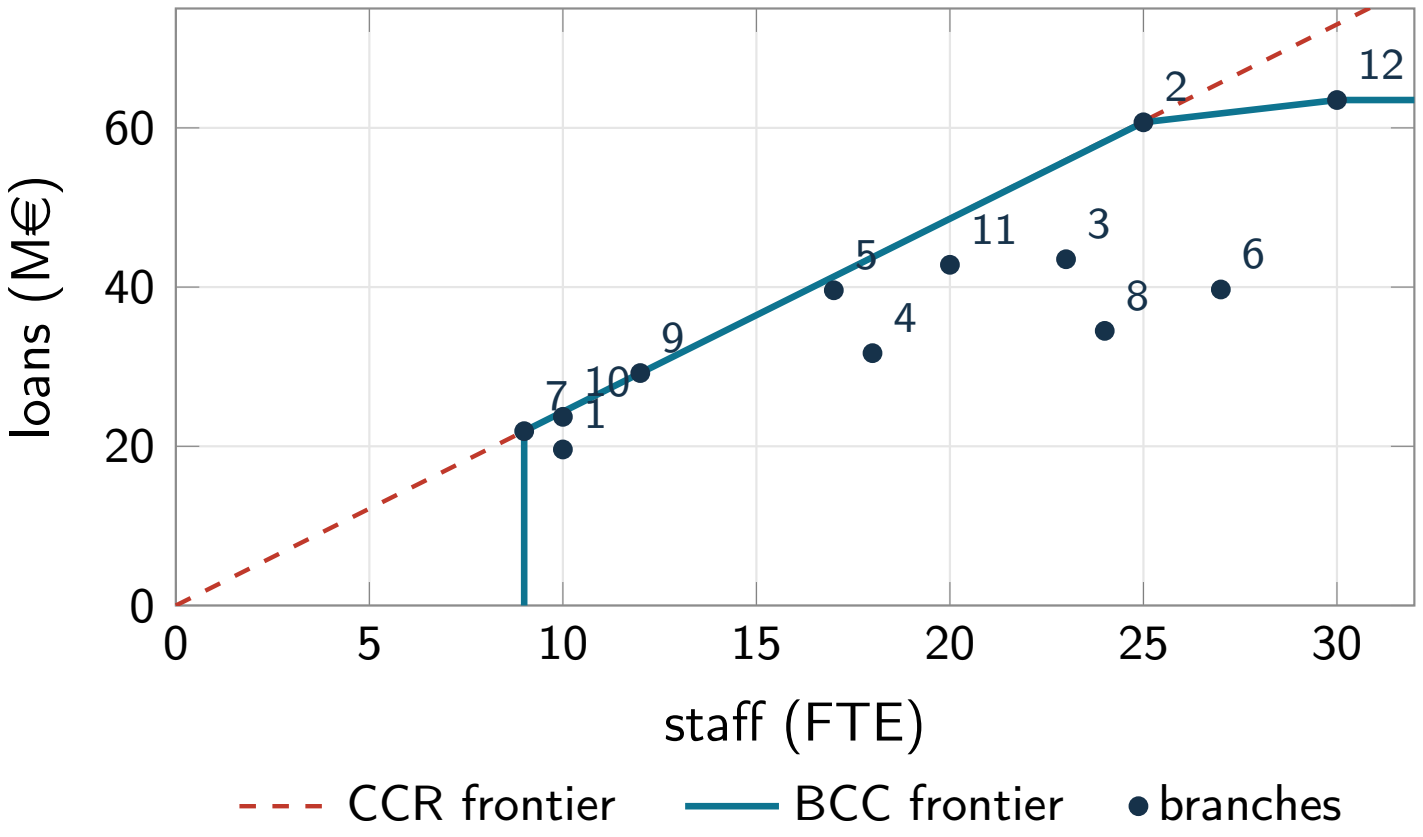
branch	$\tilde{\theta}$	peers (j: $\tilde{\lambda}_j$)	staff	target	costs	target
1	0.8055	9: 0.671	10	8.05	720	463.2
3	0.9815	7: 1.075; 10: 0.842	23	18.10	1130	1109.1
4	0.9233	7: 1.119; 10: 0.304	18	13.11	940	867.9
5	0.9604	7: 0.223; 9: 1.024; 10: 0.203	17	16.33	990	950.8
6	0.7600	7: 1.924	27	17.31	1620	1231.2
8	0.9419	7: 1.913	24	17.22	1300	1224.4
11	0.8997	2: 0.106; 10: 1.535	20	17.99	1020	917.7
12	0.9779	10: 2.679	30	26.79	1370	1339.7

- Targets after the second phase (`dea_target.csv`)

Managerial insight

- Branch 6 is the least efficient (0.76) but it is BCC-efficient: it is **too large**
- Only peer branch 7 taken 1.924 times: decreasing returns; consider splitting it into two offices
- Branch 1 is too small ($\sum \tilde{\lambda}_j = 0.671$) and inefficient also in the BCC (0.90)
- Branch 1 keeps a slack of 116.8 k€ on costs: target 463.2 against 720

Efficient frontiers: staff and loans



Reading the chart

- Reduced one-input, one-output view: the efficiencies differ from the full model
- CCR frontier: ray from the origin with slope 2.4333 M€ per FTE (branch 9)
- BCC frontier: piecewise line through branches 7, 9, 2 and 12
- Below the ray a branch pays technical and scale inefficiency; below the piecewise line only technical
- Branch 12: BCC 1, CCR 0.870 in the view

Summary: Case study and results

- Four CCR-efficient branches, eight BCC-efficient
- Branches 3, 6, 8, 12 have only scale inefficiency
- Branches 7 and 10 are the most cited peers (five times each)

Sensitivity analysis

Introduction: Sensitivity analysis

- Shadow prices of the envelopment form, checked by perturbation
- Stability of the frontier: what happens without a peer

Shadow prices: branch 6

```
branch 6: theta = 0.7600
 loans   : y = 39.7 | Pi = 0.00000 | valid in [-inf, 42.13]
 deposits: y = 73.1 | Pi = 0.01040 | valid in [68.89, inf]
Check: deposits from 73.1 to 74.1
  -> theta from 0.7600 to 0.7704 (change +0.01040)
Primal weight u = (0.00000, 0.01040)
```

- $\geq$ constraints in a minimisation: raising the right-hand side shrinks the region, $\tilde{\theta}$ does not decrease, shadow prices ≥ 0
- The shadow price of deposits is the weight $\tilde{u}_2$ of the primal

Reading the shadow prices

- One more million of deposits raises $\tilde{\theta}$ by 0.0104, for any right-hand side above 68.89
- The shadow price of loans is zero up to 42.13
- The virtual DMU already produces 2.43 million of loans more (the slack of the second phase)

Stability: without branch 7

```
  branch  3: CCR 0.9815 -> 1.0000  <- changes
  branch  4: CCR 0.9233 -> 0.9795  <- changes
  branch  5: CCR 0.9604 -> 0.9791  <- changes
  branch  6: CCR 0.7600 -> 0.9156  <- changes
  branch  8: CCR 0.9419 -> 1.0000  <- changes
Without branch 7: 5 branches change efficiency;
newly efficient: [3, 8]
```

- Efficiency is **relative**: a single DMU on the frontier changes the verdict on many
- Before acting: check the data of the most cited peers and repeat the analysis without them

Summary: Sensitivity analysis

- The shadow prices of the output constraints are the weights u_r
- Checked by perturbation, with their ranges
- The frontier depends on the observed DMUs

Exercises

Introduction: Exercises

- Five exercises to consolidate the concepts of the chapter
- Check, output orientation, weak efficiency, weight restrictions, super-efficiency
- They are solved with the laboratory script

Exercises (1/2)

- **Check.** With the optimal weights of branch 4 compute the ratio of all branches: which ones have ratio 1? How do they relate to the peers?
- **Output orientation.** Write the output-oriented CCR and check $\tilde{\varphi} = 1/\tilde{\theta}$; in the BCC the relation does not hold in general
- **A weakly efficient branch.** Branch 13 equal to branch 7 but with costs 700 k€: $\tilde{\theta}$ and slacks?

Exercises (2/2)

- **Weight restrictions.** Add $u_2 \geq u_1$ to the multiplier form: which branches remain efficient?
- **Super-efficiency.** Remove DMU o from the reference set ($\lambda_o = 0$) to rank the efficient branches: which one is the most robust?

Summary (1/2)

Model:

- Relative efficiency = weighted output/input ratio with the most favourable weights
- Charnes–Cooper: multiplier form (LP)
- Dual: envelopment form, $\tilde{\theta} = \tilde{h}_o$, peers and targets

Scale:

- BCC with $\sum_{j=1}^{n} \lambda_j = 1$; $\mathrm{SE} = \tilde{\theta}^{\mathrm{CCR}} / \tilde{\theta}^{\mathrm{BCC}}$

Summary (2/2)

Managerial message:

- Tell apart those who work badly from those who have the wrong size
- The targets say by how much to reduce each input

Caution:

- Efficiency is relative to the observed DMUs: check the most cited peers